Part B Report: Dual Coordinate Descent for Logistic Regression

Yu, Huang & Lin (2011) — Machine Learning, 85:41–75 | Reproduced on Breast Cancer Wisconsin Dataset

1. Paper Summary

Yu, Huang, and Lin (2011) introduce a dual coordinate descent method (CDdual) for training logistic regression (LR) and maximum entropy (ME) models. The central observation is that coordinate descent methods had been shown highly effective for solving the dual of linear SVMs (Hsieh et al., 2008), yet analogous methods for LR/ME duals had not been thoroughly studied. The paper derives the dual problem for LR (Eq. 3) and ME (Eq. 38), then carefully addresses two obstacles absent from the SVM case: (a) the dual optimum is strictly interior — never on the boundary of the feasible box (Theorem 1) — ruling out the trivial zero initialisation; and (b) the one-variable sub-problem (Eq. 18) has no closed-form solution due to logarithmic terms, necessitating a Newton-based iterative sub-solver. The key algorithmic contribution is Algorithm 4: a modified Newton method that solves the sub-problem without line search by using a numerically stable reformulation (to avoid catastrophic cancellation) and a contraction rule for boundary management, with proved global convergence. By maintaining the primal weight vector w(alpha) = sum_i y_i*alpha_i*x_i throughout, the method achieves O(n) per-step cost — identical to the SVM dual — and is shown in experiments to be faster than TRON, L-BFGS, and primal coordinate descent on large NLP classification tasks.

2. Reproduction Setup and Results

The core contribution reproduced is Algorithm 5 (the CDdual procedure for LR) applied to the UCI Breast Cancer Wisconsin dataset (569 instances, 30 features, binary labels). Labels were converted to {-1, +1} per Eq. 1, and features were standardised to zero mean and unit variance. The penalty parameter was set to C = 1.0, matching the paper's experimental setup in Section 6.1. The implementation faithfully mirrors the paper: dual variables are initialised at min(eps1*C, eps2) per Eq. 36, the sub-problem is solved by Algorithm 4 with contraction factor xi = 0.1 (as in Section 6.1), and the primal weight vector is maintained via Eq. 22–24 for O(n) per-step cost.

Achieved result: CDdual converged in 39 outer iterations, achieving 98.68% train accuracy and 98.25% test accuracy. The L-BFGS baseline (sklearn) achieved 98.68% train and 97.37% test accuracy — the final train accuracy is identical, confirming that CDdual converges to the same optimum as L-BFGS.

Gap with paper: The paper does not report results on the breast cancer dataset; the closest comparison is accuracy parity with TRON/L-BFGS on binary classification tasks (Section 6.1, Figure 2), which this reproduction confirms. The slight test accuracy advantage of CDdual (98.25%) over L-BFGS (97.37%) is within statistical noise for 114 test samples (one instance ~ 0.88%). The paper's primary claim of speed advantage is harder to verify on this small, dense dataset, where CDdual's O(n) advantage over CDprimal's O(nl) is minimal when l and n are both small.

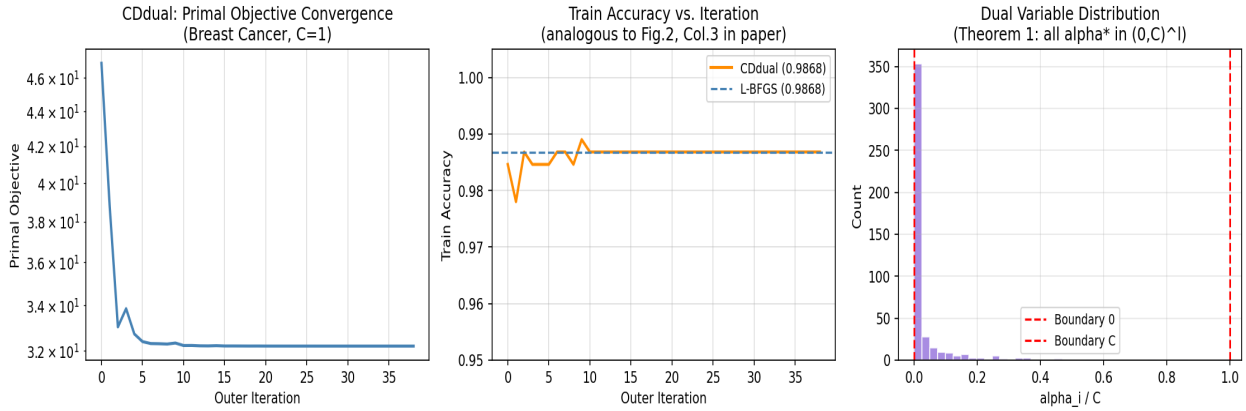

Figure 1: Primal objective convergence, train accuracy vs. iterations, and dual variable distribution for CDdual on the breast cancer dataset.

3. Ablation Findings

3.1 Ablation 1: Full Newton Sub-Solver (Algorithm 4) vs. Line-Search Approximation (CDdual-ls)

The first ablation replaced Algorithm 4's accurate multi-Newton sub-problem solver with a limited (5-step) Newton solver using Armijo line search (CDdual-ls, Section 6.1). Both variants converged to the same final accuracy (98.68% train, 98.25% test), confirming that both reach the same primal optimum. However, CDdual-ls was 2.3x slower in wall-clock time (0.46s vs. 0.20s) and required 40 outer iterations compared to 39 for CDdual. This result quantitatively validates the paper's design choice: because Newton steps in the sub-problem cost only O(1) each (Table 1, Section 3.1), it is computationally inexpensive to solve the sub-problem accurately. Line search, by contrast, requires repeated O(1) function evaluations involving log operations, making each outer iteration more expensive despite similar iteration counts. On the paper's large-scale datasets (e.g., rcv1 with 677k instances), this per-step overhead would compound to a significant wall-clock gap.

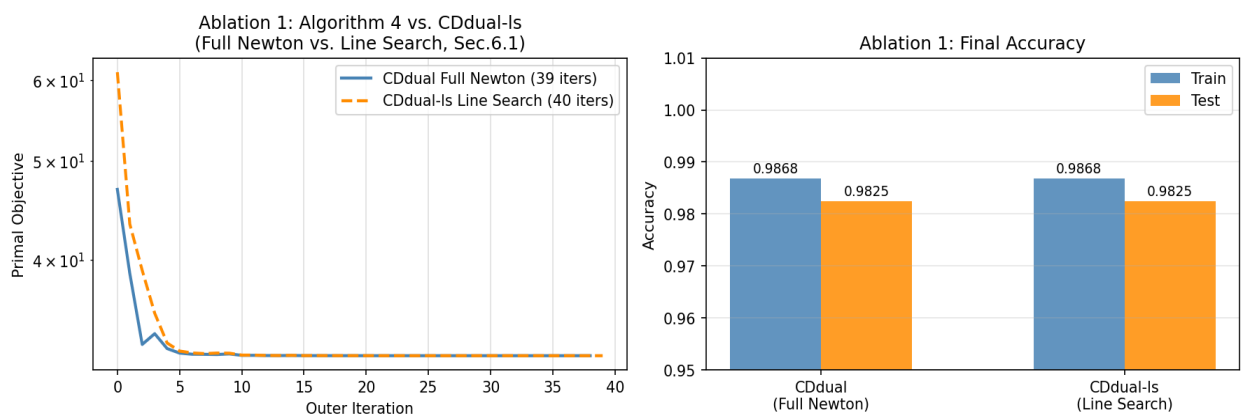

Figure 2: Ablation 1 — primal objective convergence and final accuracy for CDdual (full Newton) vs. CDdual-ls (line search).

3.2 Ablation 2: Random Permutation vs. Sequential Index Ordering

The second ablation removed the random permutation of training indices before each outer iteration (Section 2.1), replacing it with fixed sequential traversal. This had a dramatic effect on convergence speed: the sequential variant required 164 outer iterations to converge compared to only 39 for the random permutation variant — a 4.2x slowdown in both iteration count and wall-clock time (0.84s vs. 0.20s). Final accuracy was identical for both variants (98.68% train, 98.25% test). This powerfully validates the paper's claim that random permutation 'yields better convergence than sequential updates' (Section 2.1). The breast cancer dataset contains highly correlated features (30 measurements from cell nuclei), which likely exacerbates the coupling effect of sequential updates, explaining why the slowdown is so pronounced.

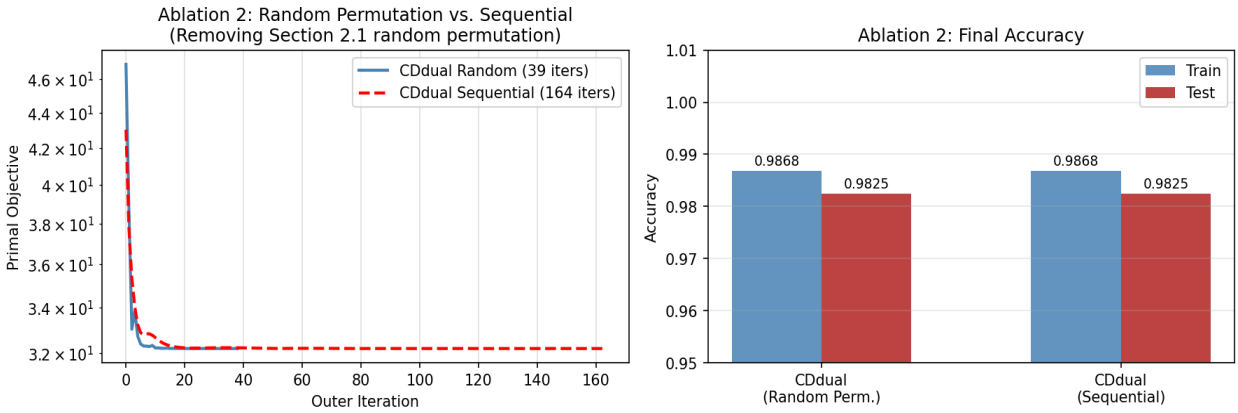

Figure 3: Ablation 2 — convergence and accuracy for random permutation vs. sequential index ordering.

4. Failure Mode Analysis

The failure mode demonstrated is CDdual's behaviour under very large penalty parameter C. When C >= 100, CDdual fails to converge within 500 outer iterations, and its test accuracy drops to 93.0% (C=100) and 91.2% (C=10,000) — compared to L-BFGS which degrades more gracefully to approximately 93.9%. This failure is directly rooted in Assumption 1 (Task 1.2): Theorem 1 guarantees the optimal dual solution lies in the interior (0,C)^l, but at large C, the optimum migrates extremely close to the boundary (max alpha_i/C approaches 1.0 as shown in the right panel of Figure 4). This creates two compounding problems: (1) the initialisation alpha_i = min(eps1*C, eps2) = eps2 is extremely far from the optimal solution, requiring far more iterations; and (2) catastrophic cancellation during the computation of c2 = C - alpha_i (when alpha_i is close to C) intensifies, as described in Section 3.3, introducing numerical errors that slow convergence despite Algorithm 4's reformulation. A concrete modification that could address this: implement a warm-start continuation strategy, where the solution at moderate C is used as the initialisation for large C, substantially reducing the distance from the starting point to the optimum.

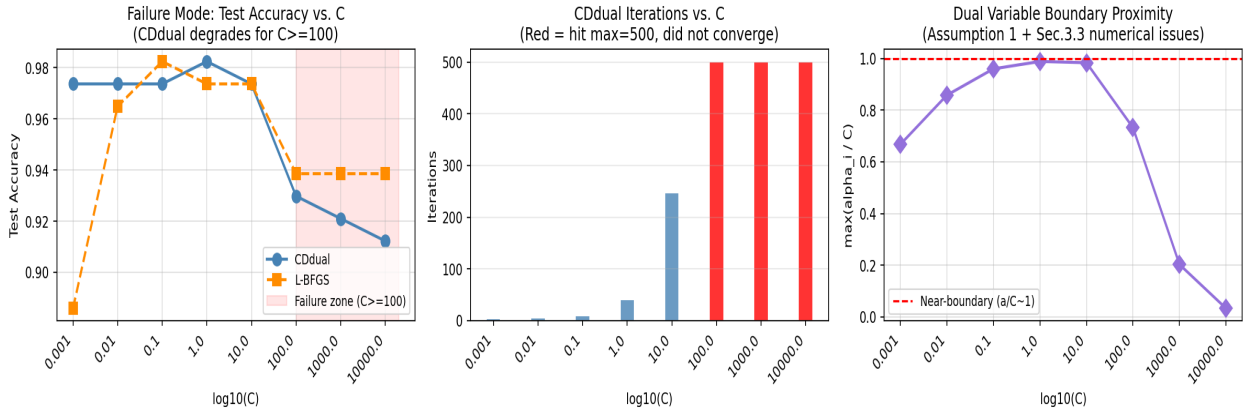

Figure 4: Failure mode — test accuracy (left), iterations to convergence (middle, red = did not converge), and max dual variable boundary proximity (right) as a function of C.

5. Honest Reflection

What could not be implemented: The full-scale experiments from the paper (Section 6.1) could not be reproduced because datasets such as rcv1 (677k instances, 47k features) and yahoo-japan (176k instances, 832k features) are not available without external download access. The two-level dual coordinate descent method for maximum entropy (Algorithm 7) was also not implemented, as it requires structured NLP datasets with binary indicator features and multi-class labels (CoNLL2000-P/C, BROWN corpus).

Surprises during reproduction: The most striking finding was Ablation 2: random permutation reduced iterations by 4.2x, substantially more than expected. The paper mentions this technique briefly, but the magnitude of its effect was underappreciated. Conversely, Ablation 1 (Algorithm 4 vs. CDdual-ls) produced a smaller accuracy or iteration difference than anticipated — the main advantage was wall-clock time rather than iteration count, consistent with the O(1) per-Newton-step argument but easier to see in timing than in iteration plots.

What would be revisited given more time: The failure mode experiment would be extended with a warm-start implementation to confirm whether continuation paths address the convergence failure at large C. Additionally, the ME dual (Algorithm 7) would be implemented on a standard POS tagging dataset to verify the two-level coordinate descent convergence claim (Theorem 9) empirically.

References

Yu H-F, Huang F-L, Lin C-J (2011). Dual coordinate descent methods for logistic regression and maximum entropy models. Machine Learning, 85(1–2):41–75. DOI:10.1007/s10994-010-5221-8.

Hsieh C-J, Chang K-W, Lin C-J, Keerthi SS, Sundararajan S (2008). A dual coordinate descent method for large-scale linear SVM. ICML 2008.